

Original Article

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Improved tip-tilt control for laser beam steering system utilizing modified disturbance observer-based controller

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Abstract A tip-tilt control system is widely utilized in a laser beam steering system (LBS) utilizing a fast steering mirror (FSM) to improve its robustness in the presence of dynamic disturbances. The ultimate goal of the LBS system involves positioning a laser beam at an aiming location point on a target with minimal beam jitter in the presence of dynamic disturbances. This paper proposes an improved tip-tilt control method utilizing a disturbance observer (DOB) in the FSM motion system. To improve tip-tilt control capability for dynamic disturbances, the existing tip-tilt control system implements a modified DOB-based control scheme, which is an improved method for disturbance suppression. Experimental results show that the control error of the modified DOB-based controller is reduced, effectively improving the disturbance attenuation capability.

1. Introduction

The tip-tilt control system utilizing a fast steering mirror (FSM) is most commonly applied to laser beam steering (LBS) systems such as near-field imaging system, laser micro scanning systems, optical free-space communications, and adaptive optics laser systems [1–8]. The control system of the LBS system must be capable of reliably aiming the laser beam at the aiming point of the target even when disturbances caused by external and internal vibrations are applied to the platform [9].

The ultimate goal for the control performance of the high-precision tip-tilt controller for an FSM system is to control the voice coil actuator or piezoelectric (PZT) actuator as per requirements under disturbance conditions. Therefore, an FSM system must be designed with a tip-tilt controller, fast response, and good disturbance attenuation for high-precision control performance [10].

The robust design of an FSM-based actuating systems is generally divided into two considerations. First, the mechanical structure must be designed to achieve the demands of the target application. FSM-based actuating systems require rigid elastic supports to satisfy the actuating forces within a required strokes range with the high-precision control [11–13]. Second, to achieve high-precision control capability, the control scheme for FSM-based actuating systems should be determined with high speed response and disturbance suppression capabilities. Gu et al. suggested eliminating vibration effects on a PZT actuator [14]. In addition, to attenuate the main resonant modes of the FSM system, Ling et al. and Wadikhaye et al. developed a notch filter as an efficient scheme [15, 16]. Furthermore, Clayton et al. and Wang et al. developed a feedforward controller [17, 18], and Schitter et al. and Gul et al. developed an input-shaping controller and integral resonant controller [19, 20], respectively.

To enhance the disturbance attenuation capability of the original control algorithm, high bandwidth and high control gain in the dynamic disturbance region are required to attenuate the disturbance sufficiently. FSM systems are widely used because the simplicity of implementation using modeled coefficients facilitates the design of high-bandwidth proportional-integral (PI) controllers [21]. The proposed disturbance observer (DOB)-based controller has been utilized

ized to motion control applications, including robot engineering, industrial automation systems, and automotive, due to its implementation simplicity and application efficiency [22-24]. In motion control, DOB-based control systems can significantly improve robustness by estimating and compensating for external disturbances and system uncertainties. In addition, adding a DOB to the original feedback system significantly improves the disturbance attenuation response because of the simplified structure and easy implementation [25, 26]. Since the DOB-based control performance is largely dependent on the mathematical model of the target plant, it is very useful to introduce proper target plant mathematical modelling and Q filter design in the FSM control system.

This paper proposes a method to improve a reliable tip-tilt control algorithm utilizing a modified DOB to enhance disturbance suppression performance at low frequencies for FSM systems. We demonstrate improved control loop performance under dynamic disturbance with a modified DOB, which is utilized to design an improved control algorithm in terms of reliability upon receiving disturbances. Therefore, this paper is organized as follows. The FSM motion system design is showcased in Sec. 2. In Sec. 3, the Modified DOB-based control scheme and its performance analyses are presented. Sec. 4 describes the verification experiments for this control scheme. Finally, the conclusions are showcased in Sec. 5.

2. FSM motion system design

2.1 PZT-based actuating module

A common approach for high-precision FSM motion systems for angular control is to utilize PZT-based actuating module. Fig. 1 illustrates the frequency characteristics of the actual plant (i.e. the PZT-based actuating module) by utilizing the dynamic signal analyzer. The PZT-based actuating module comprises a PZT-based actuator, a voltage converting driver, and a sensor signal processing unit. As shown in Fig. 1, the experimental frequency characteristics are presented for characterization of the actual plant, and the modeled plant characteristics are also shown. The PZT-based actuating module (i.e. actual plant) represents that the resonance frequency and the 5-Hz sensitivity are 425.2 Hz and 171.5 $\mu\text{rad/V}$, respectively. Further, the gains of the voltage amplification and the sensor signal amplification are 10 V/V and 0.0057 V/ μrad , respectively. The identification specifications of the actual plant are showcased in Table 1.

Further, the mathematical modeling of the nominal plant is presented below:

$$P_n(s) = \frac{K_{FSM}}{s^2 + 2\zeta\omega_n s + (\omega_n)^2} \quad (1)$$

$$= \frac{4.33 \times 10^3}{s^2 + 6.46 \times 10^3 s + 2.68 \times 10^7}$$

The PZT-based actuating module and digital signal proces-

Table 1. Identification specifications of actual plant.

Specification	Value
Resonance frequency	425.2 Hz
5-Hz sensitivity	171.5 $\mu\text{rad/V}$
Gain of voltage amplification	10 V/V
Gain of sensor signal amplification	0.0057 V/ μrad

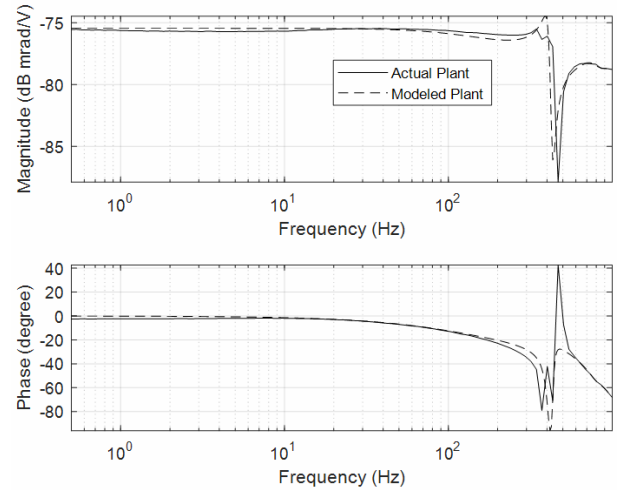


Fig. 1. Frequency response of actual plant.

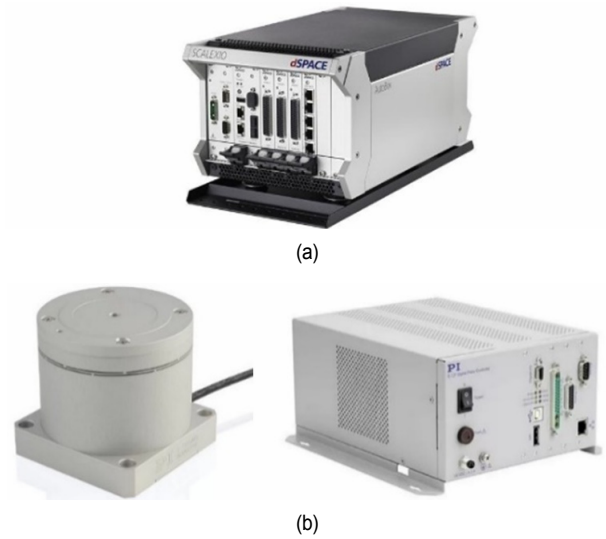


Fig. 2. Experimental setup for FSM motion system: (a) scalexio autobox; (b) S-340 module.

sor (DSP) platform utilized to implement the control capability of the FSM motion system are depicted in Figs. 2 and 3. A Scalexio AutoBox as a DSP module was utilized to control the tip-tilt actuation of the FSM motion system. In addition, a S-340 actuating module as a PZT-based actuating module was utilized within a required strokes range.

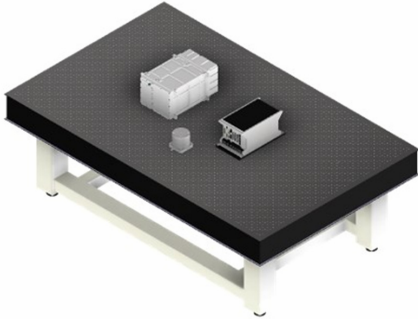


Fig. 3. Layout of FSM motion system.

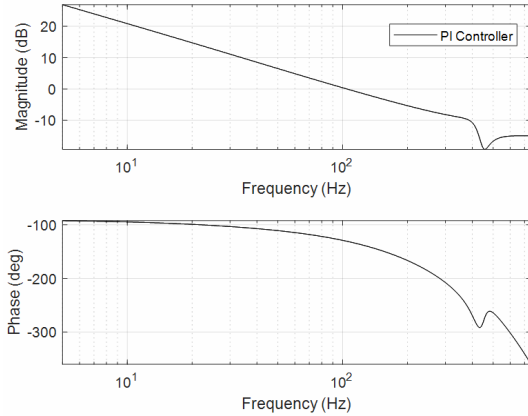


Fig. 4. Open-loop state feedback curve of designed PI controller.

2.2 System design of FSM motion system

Among the many applicable feedback controllers, including model reference adaptive control, state estimation control, stochastic robust control, and proportional-integral-derivative control, the popular used method is the proportional-integral (PI) control scheme, which is simply implementation and robust to mathematical modeling inaccuracies. Without loss of generality, the PI control scheme is designed with following mathematical modeling of the controller $C_P(s)$.

$$C_{PI}(s) = k_p + \frac{k_i}{s} \quad (2)$$

here, k_p and k_i (i.e. control gains) are the propositional and integral gains, respectively.

In general, higher control gains means better control response for the actuating command. However control gains of PI control scheme should be set appropriately the balance of the stability and capability. In this paper, to verify the control capability of PI control scheme, the gains of angular controller (i.e. PI controller) are tuned in the experimental and are chosen to be $k_p = 0.5$ and $k_i = 900$. As showcased in Fig. 4, the open-loop state feedback curve of the designed PI controller is presented. The PI control gains are designed with gain and phase margins of 6.5 dB and 49.2°, respectively, while the 0 dB

Table 2. System identifications of PI controller for FSM system.

Identification	Value
Gain margin	6.5 dB
Phase margin	49.2°
0 dB crossover frequency	105 Hz
0-Hz open loop gain	> 30 dB

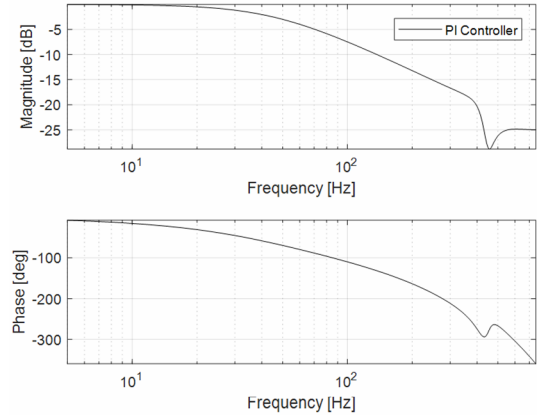


Fig. 5. Closed-loop state feedback curve of designed PI controller.

crossover frequency and 0-Hz open loop gain were 105 Hz and over 30 dB, respectively. The closed-loop state feedback curve of the designed PI controller is presented in Fig. 5. In addition, the verification experiments for this control scheme for the FSM system are indicated in Table 2.

3. Modified DOB-based control scheme

The design of a controller for a DOB-based control scheme that attempts to suppress the presence of dynamic disturbances is simplified by the ease of attaching a DOB-based control scheme to the original feedback loop, which simplifies the implementation of a DOB-based control scheme. The robustness of the DOB through mathematical modeling is achieved by adding the modeled plant uncertainty error to the original control input of the FSM angular control loop. The modeled plant uncertainty error is calculated from the nominal plant and a Q filter. Fig. 6 depicts a schematic for an FSM motion system with the DOB. The m th order Q filter is mathematically modeled as follows.

$$Q(s) = \frac{\alpha_m}{s^m + \alpha_1 s^{m-1} + \alpha_2 s^{m-2} + \dots + \alpha_m} \quad (3)$$

where m and α are the denominator order and the state coefficient of the Q filter, respectively. In general, the stability of the DOB-based control scheme is determined by the state coefficient optimization and the mathematical model order of the Q filter. Therefore, the coefficient optimization and mathematical

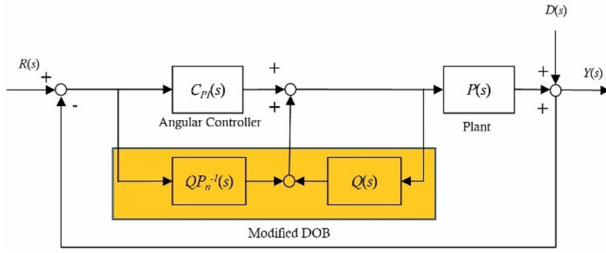


Fig. 6. Schematic of modified DOB-based control scheme for FSM motion system.

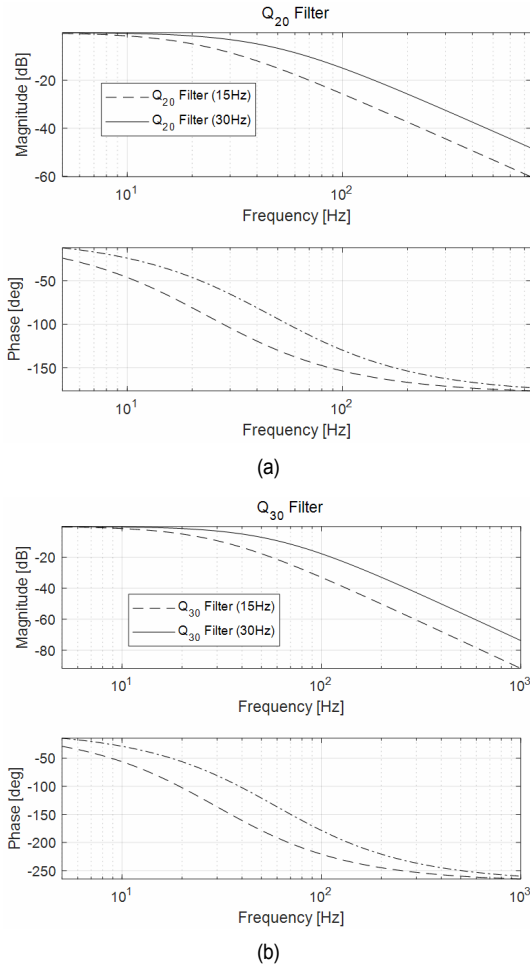


Fig. 7. Open-loop state feedback curve of modified DOB-based control scheme: (a) Q_{20} ; (b) Q_{30} filters.

model order of the Q filter should be determined manually to achieve the improvement of the robustness for the DOB-based control scheme through the experimental analysis of the noise suppression. Here, the m th denominator order Q_{20} and Q_{30} filters were utilized in the DOB-based control scheme, as depicts in Fig. 7. In a DOB-based control scheme, the stability of the control system can be predicted by analysing the control error signals (CESs), which are the measurements of an experiment. Utilizing Q_{20} and Q_{30} filters can be improved profi-

ciently the disturbance suppression capability of the FSM motion system.

In this study, Q filters were applied in low-frequency region with dynamic abilities of 15 Hz and 30 Hz to suppress the natural frequencies of mechanical vibration, which are commonly encountered in platforms equipped with laser emission sources.

As a result, the disturbance suppression capability of the DOB-based control scheme were proficiently achieved in low-frequency band with dynamic abilities of 15 Hz and 30 Hz. For dynamic abilities of 15 Hz and 30 Hz, the Q_{20} and Q_{30} filters are mathematically modeled as follows

$$Q_{20_15Hz}(s) = \frac{1}{4.6 \times 10^{-5} s^2 + 1.4 \times 10^{-2} s + 1}, \quad (4)$$

$$Q_{20_30Hz}(s) = \frac{1}{1.2 \times 10^{-5} s^2 + 6.8 \times 10^{-3} s + 1}, \quad (5)$$

$$Q_{30_15Hz}(s) = \frac{1}{4.9 \times 10^{-6} s^3 + 8.7 \times 10^{-4} s^2 + 5.1 \times 10^{-2} s + 1}, \quad (6)$$

$$Q_{30_30Hz}(s) = \frac{1}{6.1 \times 10^{-7} s^3 + 2.2 \times 10^{-4} s^2 + 2.6 \times 10^{-2} s + 1}. \quad (7)$$

As mentioned earlier, the disturbance attenuation capability of the modified DOB-based control scheme must be acceptable within the frequency region of the disturbance engaged to the application but the instability of the control scheme must be considered. By increasing the mathematical model of the filter denominator and dynamic ability of the Q filter, the disturbance attenuation capability of the modified DOB-based control scheme is better than the PI controller.

As showcased in Fig. 8, the open-loop state feedback curves with and without the modified DOB-based control scheme are as follows. The modified DOB-based control scheme is expressed mathematically modeled as follows:

$$G_{MDOB}(s) = \frac{C(s)P_n + 1 - (1 - Q(s))}{(1 - Q(s))P_n} \quad (8)$$

Fig. 8 presents open-loop state feedback curve of DOB-based control scheme. The disturbance attenuation capability of the modified DOB-based control scheme is presented in Fig. 9.

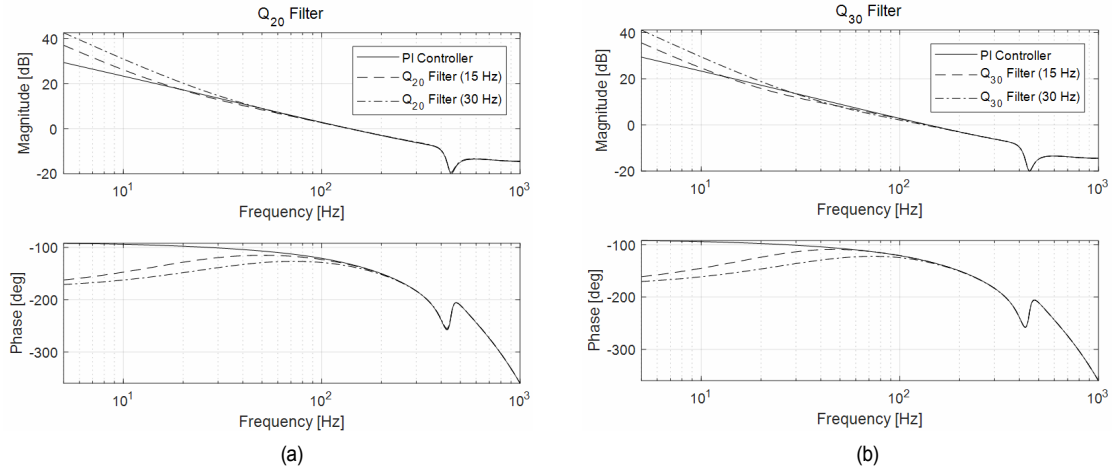
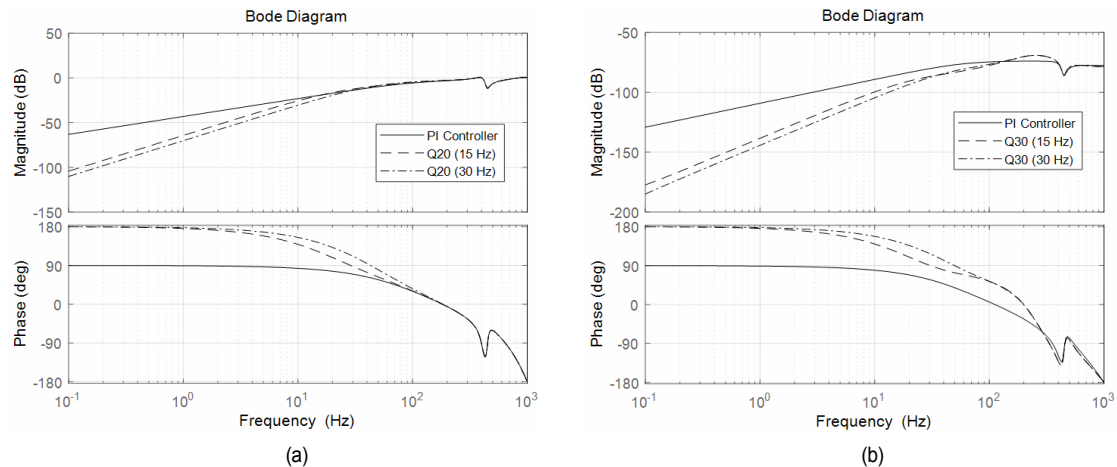
Therefore, the disturbance attenuation capability is improved through the parameter optimization of the Q filter. The dynamic ability and denominator order of the mathematical model for the Q filter can be determined through parameter optimization, which are experimental results. In this study, Q_{30} filter and Q_{20} filter are utilized, and the disturbance attenuation capability of Q_{30} filter is better than that of Q_{20} filter. To further verify the disturbance attenuation capability Q_{30} filter and Q_{20} filter, If the mathematical model of the filter denominator for the Q filters are the equal, the disturbance attenuation capability for a 30 Hz dynamic ability is better than for a 15 Hz dynamic ability, as demonstrated in Fig. 9.

Table 3. Disturbance attenuation capability for modified DOB-based control scheme (Q_{20} filter).

Specification	Tip-tilt controller (PI)	DOB-based control scheme	
		Q_{20} filter (15 Hz)	Q_{20} filter (30 Hz)
Control error (RMS)	10.34 μ rad	7.76 μ rad	6.72 μ rad
Decreased rate	-	26 %	36 %

Table 4. Disturbance attenuation capability for modified DOB-based control scheme (Q_{30} filter).

Specification	Tip-tilt controller (PI)	DOB-based control scheme	
		Q_{30} filter (15 Hz)	Q_{30} filter (30 Hz)
Control error (RMS)	10.34 μ rad	7.32 μ rad	6.41 μ rad
Decreased rate	-	30 %	39 %

Fig. 8. Open-loop state feedback curve of modified DOB-based control scheme: (a) Q_{20} ; (b) Q_{30} filters.Fig. 9. Disturbance attenuation capability of modified DOB-based control scheme: (a) Q_{20} ; (b) Q_{30} filters.

4. Modified DOB-based modified DOB-based control scheme performance

The performances of disturbance attenuation capability of modified DOB-based control scheme are presented in Figs. 9(a) and (b). The attenuation responses of the PI controller for FSM motion system with and without the DOB reveal that the target angular position was maintained successfully at 500 μ rad. Fig. 10(a) showcased that the root mean square (RMS) of the CES for the PI controller is 10.34 μ rad. When the 15-Hz-dynamic ability Q_{20} and Q_{30} filters were utilized in the FSM motion system with the DOB, the RMS of CES was 7.76 and

7.32 μ rad, respectively, as illustrated in Figs. 10(b) and (c). Furthermore, when the 30-Hz-dynamic ability Q_{20} and Q_{30} filters were employed in the FSM motion system with the DOB, the RMS of CES was 6.72 and 6.41 μ rad, respectively, as showcased in Figs. 10(d) and (e).

Eventually, the performance of the disturbance attenuation capability of the Q_{30} filter for the modified DOB-based control scheme was better than that of the Q_{20} filter. Therefore, as the mathematical model of the filter denominator and dynamic ability of the Q filter increased, the disturbance attenuation capability can be improved. The performances of the disturbance attenuation capability for the modified DOB-based control

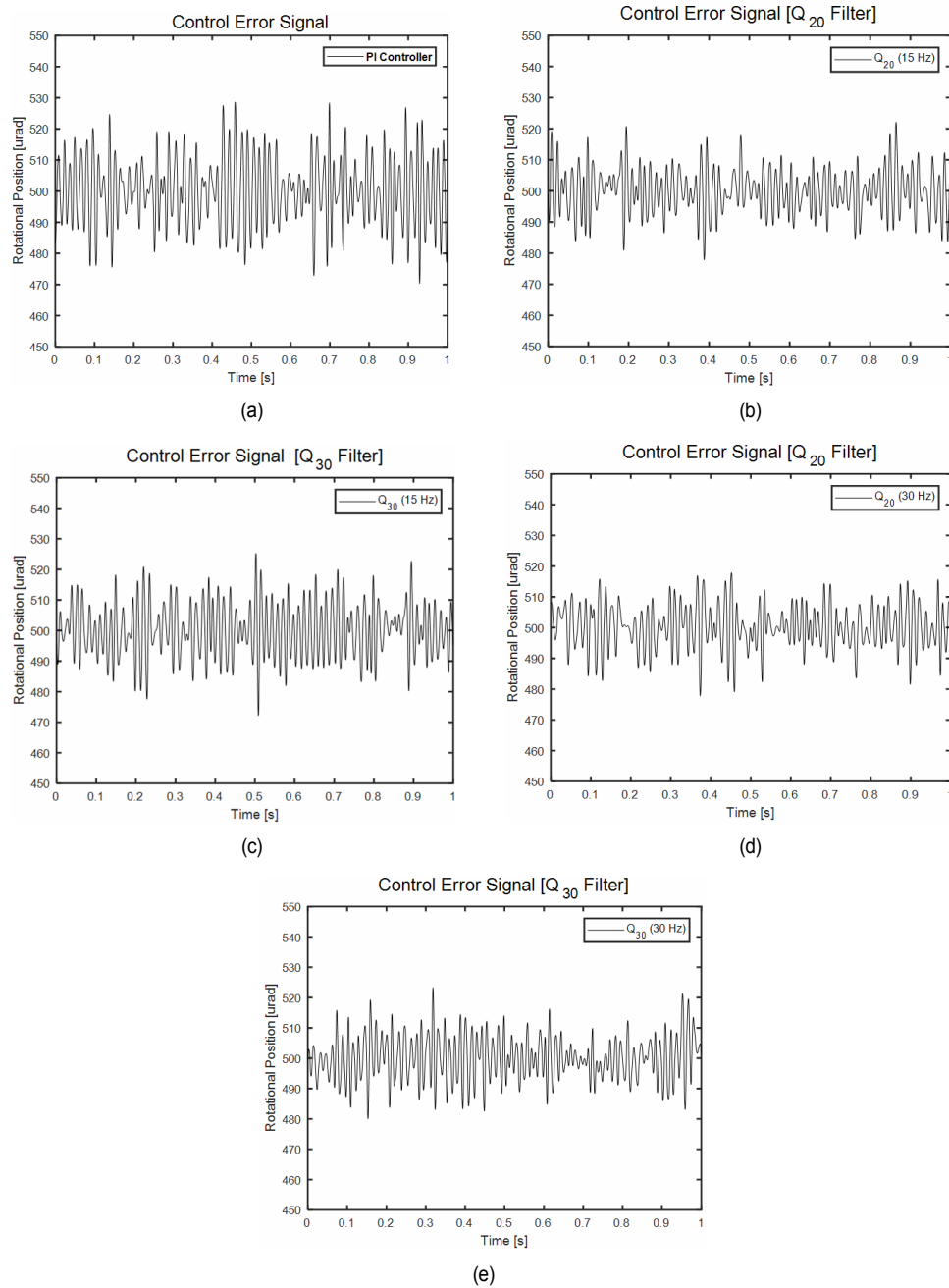


Fig. 10. Performances of disturbance attenuation capability for modified DOB-based control scheme: (a) tip-tilt controller (PI); (b) Q_{20} filter (dynamic ability: 15 Hz); (c) Q_{30} filter (dynamic ability: 15 Hz); (d) Q_{20} filter (dynamic ability: 30 Hz); (e) Q_{30} filter (dynamic ability: 30 Hz).

scheme are summarized in Tables 3 and 4.

5. Conclusion

In this study, we improved the disturbance attenuation capability of an FSM motion system by utilizing a modified DOB-based control scheme to improve the tip-tilt control performance. In the case of the 15-Hz-dynamic ability Q_{20} filter in the modified DOB-based control, the CES decreased by about 26 % compared to the case without the modified DOB-based

control scheme. Furthermore, when we utilized the 15-Hz-dynamic ability Q_{30} filter in the modified DOB-based control scheme, the CES reduced by about 30 % compared to the case without the modified DOB-based control scheme.

Furthermore, to enhance the disturbance attenuation capability of the FSM motion system with the DOB, the dynamic ability of the Q_{20} and Q_{30} filters in the modified DOB-based control scheme was set to 30 Hz. Compared to the case without the DOB, the CES was reduced by about 36 % and 39 % with the Q_{20} and Q_{30} filters. The disturbance attenuation capati-

bility of an modified DOB-based control scheme is improved via the application of the proposed control algorithm. Consequently, it is better than that observed in the original control algorithms. As part of future research, the proposed control algorithm will improve the control performance under nonlinear dynamic disturbances, which may be applicable to multi-axis laser beam steering systems.

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